

Exact Phase-Type Point Processes for Certified Measurement and Prediction of Market Event Streams

Anonymous authors

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Abstract

Temporal point processes are used both to *measure* market mechanisms, for example how fast prices absorb news and how much of the reaction is the market responding to itself, and to *predict* the next event. We show that one algebraic object, the phase-type (Erlang-chain) state space, serves both purposes with exact likelihoods. For measurement, we build a multiscale phase-type Hawkes process whose parameters all enter linearly. Its log-likelihood is concave, and we compute the global maximum together with a Fenchel duality-gap certificate. A Markov embedding gives the response to an exogenous shock, echoes included, in closed form, and this response is stable if and only if the spectral radius of the branching matrix is below one. Applied to six markets around 670 scheduled macroeconomic releases (2022–2026) with a built-in placebo test, the model shows that the direct reaction to CPI, payrolls, PPI and the FOMC statement is absorbed in about a second, while the market’s own echoes stretch half of the response to 16–137s. For prediction, we introduce EPT-X, a neural point process that adds a history-dependent mixture-of-Erlang renewal channel to a linear Hawkes backbone. Its intensity and compensator remain in closed form, so it is trained and evaluated on the exact likelihood. EPT-X has the best test likelihood on all five LOBSTER order-book streams against six classical estimators and seven neural baselines, improves the best published result on Retweet by more than 0.6 nats per event, is within one standard deviation of it on Taobao and StackOverflow, and trails the state-space model S2P2 on Taxi and Amazon. We also find that a widely used IntensityFree implementation clamps short inter-event times before scoring, which overstates its likelihood on streams with many short gaps.

1 Introduction

Price changes, order-book messages and trades arrive as streams of events in continuous time, and temporal point processes are the standard way to model them. They are used for two rather different purposes. In econometrics, Hawkes processes (Hawkes, 1971) are used to *measure* market mechanisms. Their branching structure separates exogenous events, such as a reaction to news, from endogenous events triggered by earlier trading, and the fitted kernels describe how quickly information is absorbed (Hardiman et al., 2013; Filimonov & Sornette, 2012; Rambaldi et al., 2015). In machine learning, neural temporal point processes are used to *predict* the next event, and they are compared by their held-out likelihood on shared benchmarks (Du et al., 2016; Mei & Eisner, 2017; Zuo et al., 2020; Shchur et al., 2020; Xue et al., 2024; Chang et al., 2024).

Each use has a weakness. Measurement relies on non-convex fits whose optimality is unknown, and on summaries that ignore part of the response. A common example is the “half-life” $\ln 2/\beta$ of an exponential news kernel with decay rate β , which describes the direct reaction to news but not the further trading that this reaction sets off. On the prediction side, most intensity-based neural models estimate the compensator $\int \lambda(t) dt$ by Monte Carlo (Mei & Eisner, 2017; Zuo et al., 2020; Yang et al., 2022; Chang et al., 2024), so their reported likelihoods contain sampling error. Models with an exact likelihood do exist. FullyNN (Omi

et al., 2019) models the cumulative hazard with a monotone network, and IntensityFree (Shchur et al., 2020) models the density of the next gap directly. Both, however, give up the Hawkes structure that makes a model interpretable (Shchur et al., 2021). We also find that reported likelihoods can depend on implementation details.

In this paper we show that a single algebraic object serves both purposes with exact likelihoods. We call it a *phase-type state space*: every kernel is a nonnegative combination of Erlang densities whose rates lie on a logarithmic grid of time scales. Between two events such a state evolves by an explicit matrix exponential, so intensities and their integrals are available in closed form. The class is also rich, since phase-type densities are dense in the set of densities on $\mathbb{R}_+$.

Contributions.

1. **Certified measurement (MSX).** We define a multiscale phase-type Hawkes process with marked news kernels in which every parameter enters the intensity linearly. Its log-likelihood is therefore concave. We compute the global maximum with accelerated EM followed by a barrier Newton step, and each fit comes with a Fenchel duality gap that bounds its distance from the optimum (Proposition 2).
2. **Echo-corrected responses in closed form.** Embedding the process in a linear system gives the mean response to a shock as $Ce^{At}z_0$. We prove that this system is stable exactly when the spectral radius of the branching matrix is below one (Theorem 3), and we obtain absorption times, echo amplification and price-discovery curves without simulation.
3. **Exact neural prediction (EPT-X).** We introduce a neural point process whose latent state is phase-type. It combines a linear Hawkes backbone with a history-dependent renewal channel, a mixture of Erlang distributions that also allows “no event”, added as a competing risk. The likelihood remains exact, the model contains the Hawkes backbone as a special case, and the renewal channel can approximate any conditional distribution of the next gap (Proposition 5).
4. **Benchmarks and a correction to one baseline.** EPT-X has the best likelihood on all five LOBSTER order-book streams and on Retweet, and it is within one standard deviation of the best published model on Taobao and StackOverflow. We also find that a widely used IntensityFree implementation replaces every inter-event time below 10^{-5} s by 10^{-5} s before scoring. Scored on the observed gaps, it loses its lead on two order-book streams.
5. **Application.** We study six markets around 670 scheduled macroeconomic releases between 2022 and 2026, with a placebo test built into the likelihood. The direct reaction to CPI, payrolls, PPI and the FOMC statement is absorbed in about a second, but the market’s subsequent trading stretches half of the response to 16–137 s and ninety percent of it to 5–11 minutes.

The ingredients are not new on their own. Linear-in-parameters kernel dictionaries (Bacry et al., 2015), EM algorithms for Hawkes processes (Veen & Schoenberg, 2008; Zhou et al., 2013), Erlang-chain embeddings and exact-likelihood neural point processes (Omi et al., 2019; Shchur et al., 2020) have all been studied. What we add is their combination: certified, closed-form measurement on one side, an exact neural model that contains the linear Hawkes process on the other, and experiments showing that both work.

2 Related work

Estimating Hawkes processes. Exponential and sum-of-exponential kernels are usually fitted by maximum likelihood (Bacry et al., 2015), for instance with the `tick` library (Bacry et al., 2018). Other approaches include EM algorithms that exploit the branching structure (Veen & Schoenberg, 2008), ADM4 with low-rank and sparse penalties (Zhou et al., 2013), the conditional-law (Wiener–Hopf) method (Bacry et al., 2016), cumulant matching (Achab et al., 2018) and INAR approximations (Kirchner, 2017). None of these methods returns a certificate of global optimality for multiscale kernels with delayed peaks. It is also known

that a mis-specified intraday seasonality biases estimated branching ratios upward (Filimonov & Sornette, 2015; Omi et al., 2017).

News in Hawkes models. Rambaldi et al. (2015) add an exogenous kernel for macroeconomic announcements to a Hawkes model of FX quotes, and Andersen et al. (2003) document how announcement surprises move prices. In this literature the speed of absorption is usually read off the exogenous kernel alone. We instead compute the full response, echoes included, in closed form, and we test the news kernels against matched placebo windows within the likelihood.

Neural temporal point processes. Shchur et al. (2021) give a review. RMTTP (Du et al., 2016) uses an exponential intensity between events, whose integral is closed form. NHP (Mei & Eisner, 2017), SAHP (Zhang et al., 2020), THP (Zuo et al., 2020), AttNHP (Yang et al., 2022) and the state-space model S2P2 (Chang et al., 2024) compute the intensity with recurrent, attention or state-space encoders, and they estimate the compensator by Monte Carlo or quadrature. FullyNN (Omi et al., 2019) models the cumulative hazard with a monotone network, and IntensityFree (Shchur et al., 2020) models the density of the next gap with a mixture of log-normals. Both have exact likelihoods but no Hawkes structure. Recent work also explores Mamba-style encoders (Gao et al., 2024) and interpretable latent Hawkes dynamics (Boyd et al., 2025). EasyTPP (Xue et al., 2024) provides shared implementations, and Bosser & Ben Taieb (2023) study how accurately neural models predict the next event.

EPT-X differs from these models in two ways. First, between events its intensity is a linear read-out of a phase-type system that can be solved exactly, plus bounded hazard terms and a renewal channel whose survival function is a mixture of Erlang survival functions. All of these integrate in closed form, so the likelihood we train on and report is exact. Second, its backbone is the linear Hawkes process itself, so a fitted model still has branching ratios and excitation kernels that can be read off.

Phase-type distributions. Mixtures of Erlang distributions are dense in the set of distributions on $(0, \infty)$ (Tijms, 1994; Lee & Lin, 2010). We use this fact for the renewal channel, and we use the Erlang-chain recursion for every state in the paper.

3 Phase-type state spaces

A dictionary of Erlang densities. Fix a set of rates $\beta_1, \dots, \beta_K > 0$ on a logarithmic grid, so that each rate corresponds to a time scale $1/\beta_k$, and an Erlang order $R \geq 1$. For $k = 1, \dots, K$ and $r = 0, \dots, R - 1$ we define the basis densities

$$g_{kr}(t) = \beta_k \frac{(\beta_k t)^r}{r!} e^{-\beta_k t}, \quad t \geq 0. \quad (1)$$

Each g_{kr} is the density of a gamma (Erlang) distribution with shape $r + 1$ and rate β_k , so $\int_0^\infty g_{kr}(t) dt = 1$ and its mean is $(r + 1)/\beta_k$. A kernel is a nonnegative combination

$$\phi(t) = \sum_{k=1}^K \sum_{r=0}^{R-1} a_{kr} g_{kr}(t), \quad a_{kr} \geq 0. \quad (2)$$

Because each basis density integrates to one, the total mass of the kernel, which is its branching ratio, is simply $\sum_{k,r} a_{kr}$. With $R = 1$ the dictionary contains only exponentials, and mixtures of exponentials can approximate any completely monotone kernel, power laws included (Bernstein’s theorem). Orders $r \geq 1$ add kernels that start at zero and peak later, which describe delayed reactions. More generally, phase-type densities are dense in the set of densities on $\mathbb{R}_+$.

Exact recursion. Every state in this paper is built from such Erlang chains: the Hawkes excitation, the response to news, the neural latent state of EPT and the renewal channel of EPT-X. For all of them, the evolution between two events and its integral are available in closed form. The following lemma gives the formulas for one source and one rate.

Lemma 1 (Erlang-chain recursion). *Let $t_1 < t_2 < \dots$ be the event times of one source, let $\beta > 0$ be one rate of the dictionary, and define the chain of states*

$$s_r(t) = \sum_{n: t_n < t} g_r(t - t_n), \quad r = 0, \dots, R-1,$$

where g_r is the basis density (1) with rate β . Write $\mathbf{s} = (s_0, \dots, s_{R-1})^\top$, let I be the $R \times R$ identity and let B be the $R \times R$ down-shift matrix ($B_{r,r-1} = 1$ and all other entries zero).

1. Between events, $\dot{\mathbf{s}} = \beta(B - I)\mathbf{s}$; at an event, s_0 increases by β .
2. For a gap $\Delta \geq 0$ with no event in $(t, t + \Delta)$, let $y = \beta\Delta$ and $w_j = e^{-y}y^j/j!$. Then

$$s_r(t + \Delta) = \sum_{j=0}^r w_j s_{r-j}(t), \quad \int_t^{t+\Delta} s_r(u) du = \frac{1}{\beta} \sum_{j=0}^r P(j+1, y) s_{r-j}(t),$$

where $P(a, y)$ is the regularised lower incomplete gamma function.

Proof. Step 1 (dynamics). Differentiating (1) gives

$$\frac{d}{dt} g_0(t) = -\beta g_0(t), \quad \frac{d}{dt} g_r(t) = \beta(g_{r-1}(t) - g_r(t)) \quad (r \geq 1).$$

Summing over past events gives $\dot{s}_0 = -\beta s_0$ and $\dot{s}_r = \beta(s_{r-1} - s_r)$, which is $\dot{\mathbf{s}} = \beta(B - I)\mathbf{s}$. Since $g_0(0) = \beta$ and $g_r(0) = 0$ for $r \geq 1$, a new event adds β to s_0 and leaves the other states unchanged.

Step 2 (evolution over a gap). The solution is $\mathbf{s}(t + \Delta) = e^{\beta(B-I)\Delta}\mathbf{s}(t)$. The matrices B and I commute and $B^R = 0$, so

$$e^{\beta(B-I)\Delta} = e^{-y} e^{yB} = e^{-y} \sum_{j=0}^{R-1} \frac{y^j}{j!} B^j.$$

Because B^j shifts a vector down by j places, component r of $e^{\beta(B-I)\Delta}\mathbf{s}(t)$ is $\sum_{j \leq r} w_j s_{r-j}(t)$.

Step 3 (integral over a gap). Integrating the weights over the gap,

$$\int_0^\Delta e^{-\beta u} \frac{(\beta u)^j}{j!} du = \frac{1}{\beta} P(j+1, y),$$

which gives the second formula. $\square$

Evaluating a likelihood therefore costs $O(NdKR^2)$ operations for N events in d dimensions, and it involves no Monte Carlo or quadrature error.

4 MSX: certified measurement with echo-corrected responses

Model. We observe d event types (for example upward and downward price moves of several assets) inside observation windows w , each with a likelihood interval $[t_0^w, t_1^w]$. Exogenous shocks e , here scheduled macroeconomic releases, occur at known times τ_e . Each shock has a type c_e (CPI, payrolls, and so on) and a nonnegative mark vector $\mathbf{z}_e = (1, z_e^+, z_e^-)$, where z_e^+ and z_e^- are the positive and negative parts of the standardised surprise. The MSX intensity of event type i is

$$\begin{aligned} \lambda_i(t) = & \underbrace{c_{iw} + \sum_m \eta_{im} h_m(\text{tod}(t))}_{\text{baseline}} + \underbrace{\sum_{j=1}^d \sum_{n: t_n^j < t} \sum_{k,r} a_{ijk r} g_{kr}(t - t_n^j)}_{\text{endogenous}} \\ & + \underbrace{\sum_{e: \tau_e < t} \sum_{k,r,m} b_{ic_e k r m} z_{em} g_{kr}^{\text{exo}}(t - \tau_e)}_{\text{response to news}} + \underbrace{\sum_{e: \tau_e > t} \sum_k \alpha_{ic_e k} \beta_k^{\text{ant}} e^{-\beta_k^{\text{ant}}(\tau_e - t)}}_{\text{anticipation}}. \end{aligned} \quad (3)$$

The terms are as follows.

- *Baseline.* c_{iw} is an intercept for window w , and h_m are periodic hat functions of the local time of day $\text{tod}(t)$ with weights η_{im} .
- *Endogenous excitation.* t_n^j is the n -th event of type j , the g_{kr} are the basis densities (1), and a_{ijk_r} is the weight with which an event of type j excites type i at time scale k and order r .
- *Response to news.* g_{kr}^{exo} is a second dictionary with its own rates, and b_{ickrm} weights mark component m of a release of type c .
- *Anticipation.* α_{ick} captures activity that builds up before a release of type c , with rates β_k^{ant} .

All parameters are nonnegative. Every term is linear in its parameters, so we can write $\lambda_i(t) = \mathbf{x}(t)^\top \boldsymbol{\theta}_i$, where $\boldsymbol{\theta}_i$ stacks the parameters of type i and $\mathbf{x}(t) \geq 0$ stacks the corresponding known functions of time. The log-likelihood then separates across event types:

$$\ell_i(\boldsymbol{\theta}_i) = \sum_{n: u_n=i} \log(\mathbf{x}(t_n)^\top \boldsymbol{\theta}_i) - \boldsymbol{\Phi}^\top \boldsymbol{\theta}_i, \quad \boldsymbol{\Phi} = \sum_w \int_{t_0^w}^{t_1^w} \mathbf{x}(t) dt, \quad (4)$$

where u_n is the type of the n -th event. Every entry of $\mathbf{x}(t_n)$ and of $\boldsymbol{\Phi}$ follows from Lemma 1.

Concavity and a certificate. Problem (4) maximises a concave function over the nonnegative orthant. We add an ℓ_1 penalty $\gamma^\top \boldsymbol{\theta}_i$ with weights $\gamma \geq 0$ proportional to each feature’s exposure; since $\boldsymbol{\theta}_i \geq 0$, the penalty is linear and simply adds γ to $\boldsymbol{\Phi}$. We maximise with the multiplicative EM update

$$\theta_p \leftarrow \theta_p \frac{\sum_n x_{np}/\lambda_n}{\Phi_p + \gamma_p},$$

where x_{np} is feature p at the n -th event of type i and λ_n the current intensity there. We accelerate it with SQUAREM (Varadhan & Roland, 2008) and finish with a log-barrier Newton method. The following bound tells us how far any fit is from the optimum.

Proposition 2 (Duality-gap certificate). *Fix an event type and drop the index i . Let X be the matrix whose n -th row is $\mathbf{x}(t_n)^\top$ for the N events of this type, let $\boldsymbol{\theta} \geq 0$ be any parameter vector with intensities $\lambda_n = (X\boldsymbol{\theta})_n > 0$, and let ℓ^* be the maximum of the penalised log-likelihood $\ell(\boldsymbol{\theta}) = \sum_n \log \lambda_n - (\boldsymbol{\Phi} + \boldsymbol{\gamma})^\top \boldsymbol{\theta}$. Define*

$$s = \max_p \frac{\sum_n x_{np}/\lambda_n}{\Phi_p + \gamma_p}.$$

Then

$$\ell^* - \ell(\boldsymbol{\theta}) \leq N \log s + (\boldsymbol{\theta}^\top (\boldsymbol{\Phi} + \boldsymbol{\gamma}) - N),$$

and both terms on the right are zero at the optimum.

The proof, in Appendix A, applies Fenchel’s inequality to each $\log \lambda_n$. Every fit reported in this paper comes with this bound.

Markov embedding. To study the response to a single release, we collect all endogenous Erlang states s_{jkr} into one vector $\mathbf{u}(t)$ and the states of the release’s news kernels into $\mathbf{v}(t)$. Let F be the block-diagonal generator of the endogenous chains (one block $\beta_k(B - I)$ per source and rate, as in Lemma 1) and F_e the generator of the news chains. Let G_{in} be the matrix that adds β_k to the first phase of every chain of source j when an event of type j occurs. Finally, let Θ hold the endogenous weights a_{ijk_r} and Θ_e the news weights of this release, so that the excess intensity is $\Theta \mathbf{u}(t) + \Theta_e \mathbf{v}(t)$.

Taking expectations, each event of type j occurs at the mean rate $m_j(t)$ and adds $G_{\text{in}} e_j$ to $\mathbf{u}$. The mean excess intensity $\mathbf{m}(t)$ caused by the release therefore satisfies a linear system:

$$\dot{\mathbf{z}} = \mathcal{A} \mathbf{z}, \quad \mathcal{A} = \begin{pmatrix} F + G_{\text{in}} \Theta & G_{\text{in}} \Theta_e \\ 0 & F_e \end{pmatrix}, \quad \mathbf{m}(t) = C \mathbf{z}(t), \quad C = \begin{pmatrix} \Theta & \Theta_e \end{pmatrix}, \quad (5)$$

where $\mathbf{z} = (\bar{\mathbf{u}}, \mathbf{v})$ stacks the expected endogenous states and the news states, and $\mathbf{z}(0) = \mathbf{z}_0 = (\mathbf{0}, \mathbf{v}_0)$ with $\mathbf{v}_0$ the news states just after the release.

Theorem 3 (Stability). *Let G be the $d \times d$ branching matrix with entries $G_{ij} = \sum_{k,r} a_{ijk_r}$, the expected number of type- i events triggered directly by one type- j event, and let $\rho(G)$ be its spectral radius. Then the matrix $F + G_{\text{in}}\Theta$ is Hurwitz (all eigenvalues have negative real part) if and only if $\rho(G) < 1$.*

Proof. Step 1. Every block $\beta_k(B - I)$ of F is lower triangular with diagonal $-\beta_k < 0$ and nonnegative off-diagonal entries. Hence F is Hurwitz with nonnegative off-diagonal entries, and $-F$ is a nonsingular M-matrix. The matrix $P = G_{\text{in}}\Theta$ is entrywise nonnegative.

Step 2. For such a pair, $F + P$ is Hurwitz if and only if $\rho((-F)^{-1}P) < 1$. This is the regular-splitting theorem for M-matrices (Varga, 2000, Thm. 3.29), applied to $-(F + P) = (-F) - P$.

Step 3. The products AB and BA have the same nonzero eigenvalues, so

$$\rho((-F)^{-1}G_{\text{in}}\Theta) = \rho(\Theta(-F)^{-1}G_{\text{in}}).$$

Step 4. Column j of $(-F)^{-1}G_{\text{in}} = \int_0^\infty e^{Ft}G_{\text{in}} dt$ contains, for every chain of source j , the integral over time of the state left by a single type- j event. By Lemma 1 this state is $g_{kr}(t)$, whose integral is one. Hence

$$(\Theta(-F)^{-1}G_{\text{in}})_{ij} = \sum_{k,r} a_{ijk_r} \int_0^\infty g_{kr}(t) dt = \sum_{k,r} a_{ijk_r} = G_{ij}.$$

Combining Steps 2 to 4, $F + G_{\text{in}}\Theta$ is Hurwitz if and only if $\rho(G) < 1$. □

Corollary 4 (Closed-form absorption). *Assume $\rho(G) < 1$. Then $\mathcal{A}$ is Hurwitz, because it is block triangular and F_e is Hurwitz, and the following quantities are available in closed form.*

1. The mean excess intensity after a release, echoes included: $\mathbf{m}(t) = Ce^{At}\mathbf{z}_0$.
2. Its total, the expected number of extra events: $M = \int_0^\infty \mathbf{m}(t) dt = -CA^{-1}\mathbf{z}_0$.
3. The fraction of the response absorbed by time t , summed over event types:

$$\Pi(t) = \frac{\mathbf{1}^\top CA^{-1}(e^{At} - I)\mathbf{z}_0}{-\mathbf{1}^\top CA^{-1}\mathbf{z}_0}.$$

Its quantiles $t_q = \Pi^{-1}(q)$ are the echo-corrected absorption times; we report t_{50} and t_{90} .

4. The amplification $\kappa = M/M_{\text{direct}}$, where $M_{\text{direct}} = -\Theta_e F_e^{-1}\mathbf{v}_0$ is the total of the direct news response $\mathbf{m}_{\text{direct}}(t) = \Theta_e e^{F_e t}\mathbf{v}_0$ alone.
5. For an asset whose price is observed through upward and downward crossings of size δ (types u and v), the expected price change up to time t : $\delta \int_0^t (m_u(s) - m_v(s)) ds$.

The usual “news half-life” $\ln 2/\beta$ describes only $\mathbf{m}_{\text{direct}}$. When $\rho(G)$ is close to one, the echoes dominate the response, and Section 7 shows that the two can differ by one to two orders of magnitude.

5 EPT-X: exact neural prediction

We now turn to prediction. Events arrive at times $t_1 < t_2 < \dots$ with marks $y_n \in \{1, \dots, M\}$, and we write $\Delta_n = t_n - t_{n-1}$ for the gaps. For a time t after the n -th event, $s = t - t_n$ denotes the time elapsed since that event.

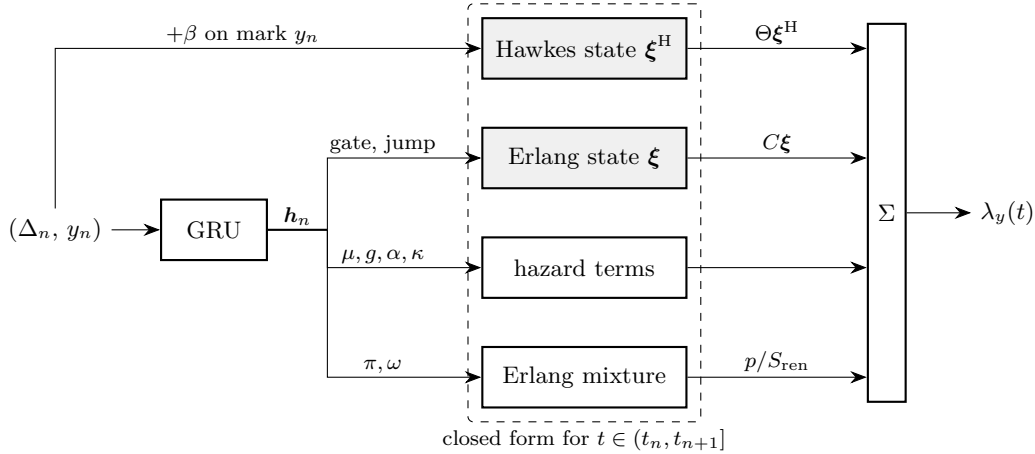


Figure 1: EPT-X at one event. The GRU summary $\mathbf{h}_n$ sets the jump of the neural Erlang state, the hazard parameters and the renewal weights, while the Hawkes state is driven by the event mark alone. The grey boxes are the states carried from one event to the next. Until the next event, all four terms evolve in closed form, and their sum is the intensity $\lambda_y(t)$.

EPT: a neural phase-type state. The latent state $\boldsymbol{\xi}(t) \in \mathbb{R}_{\geq 0}^P$ consists of L channels of Erlang chains of order R , each with its own learnable rate. Between events it evolves exactly as in Lemma 1. At each event, a recurrent network summarises the history: a GRU (Cho et al., 2014) receives an embedding of the mark y_n and log transforms of the gap Δ_n and produces a hidden vector $\mathbf{h}_n$. In EPT-X the gap enters through its standardised logarithm, which keeps nanosecond-scale gaps distinguishable from zero. The state then jumps to

$$\boldsymbol{\xi}(t_n^+) = \sigma(W_g \mathbf{h}_n) \odot \boldsymbol{\xi}(t_n^-) + \text{softplus}(W_\xi \mathbf{h}_n),$$

where σ is the logistic function, $\odot$ the elementwise product and W_g, W_ξ learned matrices. The first term lets the network forget part of the old state; the second adds a nonnegative jump. For $t \in (t_n, t_{n+1}]$ the intensity of mark y is

$$\lambda_y^{\text{EPT}}(t) = \mu_y + \sum_{p=1}^P C_{yp} \xi_p(t) + g_{1y} e^{-\alpha s} + g_{2y} (1 - e^{-\kappa s}) + \sum_q \Theta_{yq} \xi_q^{\text{H}}(t). \quad (6)$$

The quantities $\mu_y, C_{yp}, g_{1y}, g_{2y}, \alpha$ and κ are nonnegative outputs of the network computed from $\mathbf{h}_n$, so they stay fixed until the next event. The two hazard terms let the intensity decay or rise with s , but stay bounded. An unbounded Gompertz term would overflow on heavy-tailed gaps. The last term is a *linear multivariate phase-type Hawkes process*: $\xi^{\text{H}}(t)$ collects Erlang chains on a fixed logarithmic grid of rates, one group per source mark, and every event of mark y adds β_k to the first phase of the chains of source y . Its weights $\Theta_{yq} \geq 0$ are learned. This term is a residual path: when the neural read-outs are zero, EPT is the Hawkes process of Section 4, so the network only has to learn departures from linear excitation.

The renewal channel. Order-book and social streams contain cascades, several events in a very short interval followed by a pause. A linear Hawkes intensity describes excitation *between* cascades well, but it has no direct handle on the distribution of gaps *within* them. EPT-X therefore adds a second channel that models the next gap directly.

We fix Q Erlang “atoms” indexed by q , each with an order R_q and a rate β_q . Their means R_q/β_q lie on a logarithmic grid that spans the time scales of the data, and their orders come from $\{1, 4, 16\}$, optionally up to 1024. An atom of order R_q has coefficient of variation $R_q^{-1/2}$, so high orders give sharp peaks in log-time. From $\mathbf{h}_n$ the network outputs:

- weights $\pi_1, \dots, \pi_Q, \pi_\infty \geq 0$ summing to one, where π_∞ is the weight of a *defect* atom meaning “the renewal channel produces no event”;

- for each time scale, a distribution $\omega_{y|q}$ over the marks, so that the mark of a burst may depend on how quickly it follows the last event;
- optionally, a bounded shift of each log-rate, which lets an atom move within its grid cell.

The renewal channel’s joint density of the next gap s and mark y , and its survival function, are

$$p(s, y) = \sum_{q=1}^Q \pi_q \omega_{y|q} \frac{\beta_q^{R_q} s^{R_q-1} e^{-\beta_q s}}{(R_q - 1)!}, \quad S_{\text{ren}}(s) = \pi_\infty + \sum_{q=1}^Q \pi_q e^{-\beta_q s} \sum_{i=0}^{R_q-1} \frac{(\beta_q s)^i}{i!}. \quad (7)$$

EPT-X adds the corresponding hazard to EPT:

$$\lambda_y(t) = \lambda_y^{\text{EPT}}(t) + \frac{p(s, y)}{S_{\text{ren}}(s)}.$$

Figure 1 shows how the parts fit together. The network only acts at event times; between events every term evolves in closed form.

Proposition 5. *Let $\Lambda^{\text{EPT}}(\Delta) = \int_{t_n}^{t_n+\Delta} \sum_y \lambda_y^{\text{EPT}}(t) dt$ be the compensator of EPT over a gap of length Δ .*

1. *The compensator of EPT-X over the same gap is $\Lambda^{\text{EPT}}(\Delta) - \log S_{\text{ren}}(\Delta)$, and both terms are in closed form.*
2. *If $\pi_\infty = 1$, EPT-X coincides with EPT, which in turn contains the linear phase-type Hawkes process.*
3. *Given $\mathbf{h}_n$, the gap produced by the renewal channel is the absorption time of a finite continuous-time Markov chain, so its survival function, quantiles and moments are closed-form matrix expressions.*
4. *As the grid of atoms is refined, the renewal channel can approximate any conditional distribution of the next gap.*

Proof. Part 1. The intensity of EPT-X is a sum of two intensities, so its compensator is the sum of the two compensators. For the renewal channel, $\sum_y p(s, y) = -\frac{d}{ds} S_{\text{ren}}(s)$, because each Erlang density is the negative derivative of its survival function. Hence

$$\int_0^\Delta \frac{\sum_y p(s, y)}{S_{\text{ren}}(s)} ds = - \int_0^\Delta \frac{d}{ds} \log S_{\text{ren}}(s) ds = - \log S_{\text{ren}}(\Delta),$$

using $S_{\text{ren}}(0) = 1$. The EPT part is closed form by Lemma 1 and by direct integration of the two exponential hazard terms.

Part 2. If $\pi_\infty = 1$, all other weights are zero. Then $p \equiv 0$ and $S_{\text{ren}} \equiv 1$, so the renewal hazard vanishes. Setting the neural read-outs of EPT to zero leaves the Hawkes term.

Part 3. An Erlang distribution of order R is the absorption time of a chain of R exponential phases. A mixture of such chains, entered at random according to the weights π_q , together with an absorbing “never” state of weight π_∞ , is again a finite continuous-time Markov chain with one absorbing state.

Part 4. Mixtures of Erlang distributions are dense in the distributions on $(0, \infty)$ (Tijms, 1994; Lee & Lin, 2010). The renewal channel is such a mixture, with weights that depend on the history. This is the same expressiveness as a conditional mixture density (Shchur et al., 2020), with the difference that the intensity and compensator stay exact. $\square$

Training and prediction. We maximise the exact log-likelihood $\sum_n \log \lambda_{y_n}(t_n) - \Lambda$, where Λ is the total compensator, so the gradients contain no sampling noise. For large orders we evaluate S_{ren} through the regularised upper incomplete gamma function, and we take care that its gradient stays finite at zero gaps. The log-likelihood splits exactly into a time term, $\log \lambda(t_n) - \Lambda$ with $\lambda = \sum_y \lambda_y$, and a mark term, $\log(\lambda_{y_n}(t_n)/\lambda(t_n))$, which we report separately. Predictions of the next event are Bayes-optimal functionals of the exact density: the expected gap is $\int_0^\infty S(s) ds$, where S is the survival function of the full model, and the predicted mark is $\arg \max_y \int_0^\infty \lambda_y(s) S(s) ds$. Both integrals are computed by one-dimensional quadrature on a logarithmic grid.

6 Experiments

All configuration choices are made on validation data, and the test split is scored once.

6.1 Synthetic recovery

We simulate five ground truths by exact cluster sampling. Several of them lie deliberately outside the MSX class:

- S1: a single exponential kernel;
- S2: a power-law (Lomax) kernel with tail exponent 1.5;
- S3: a gamma-shaped delayed peak plus a fast exponential;
- S4: a sparse four-dimensional network whose kernels have components at 10 ms and 30 s;
- S5: a sparse ten-dimensional network of power-law kernels.

Each scenario has ten windows per split and five seeds. We compare MSX with the exponential and sum-of-exponential maximum-likelihood estimators, the nonparametric EM, ADM4, the conditional-law method and NPHC. MSX-auto chooses its Erlang order and ℓ_1 level on held-out training windows.

Table 1 reports how far each method’s held-out log-likelihood falls below that of the true model. When the truth is a single exponential, the estimators that assume exactly this form are best. Under a power law, the sum-of-exponential estimators and MSX perform about equally. When the kernel has structure that no single family captures (a delayed peak, two separated time scales or a power-law network), MSX-auto has the smallest deficit, 7 to 10 times smaller than the best competitor’s. On the two network scenarios it also roughly halves the best competitor’s branching-matrix error and recovers the causal graph perfectly (AUC = 1, as do several competitors). On the delayed peak, the nonparametric EM estimates the branching matrix slightly better (Table 9). Over all 100 MSX fits, the certificate of Proposition 2 was at most 3.3×10^{-4} nats.

Table 1: Synthetic recovery: held-out log-likelihood deficit to the true model (10^{-3} nats per event, lower is better; mean $\pm$ s.d. over five seeds). Bold: best; underlined: second best; shaded: our estimators.

	S1 exp	S2 power law	S3 hump	S4 multiscale	S5 network
<i>Classical estimators</i>					
Exp. kernel MLE	0.05 ± 0.05	26.05 ± 0.50	101.47 ± 1.81	711.28 ± 53.09	426.80 ± 368.57
Sum-of-exp. MLE	0.92 ± 0.13	0.18 ± 0.12	11.51 ± 0.46	—	—
ADM4	<u>0.05</u> ± 0.05	26.05 ± 0.50	101.47 ± 1.81	3.13 ± 0.25	26.51 ± 0.29
Hawkes EM (nonparam.)	1.02 ± 0.07	1.73 ± 0.21	2.24 ± 0.12	4.55 ± 0.29	5.23 ± 0.25
Conditional law	2.84 ± 1.03	4.91 ± 0.59	10.66 ± 2.13	114.94 ± 1.20	78.64 ± 4.79
<i>Ours</i>					
MSX ($R = 2$)	0.23 ± 0.12	0.32 ± 0.12	5.90 ± 0.36	0.54 ± 0.12	1.19 ± 0.07
MSX ($R = 4$)	0.28 ± 0.15	0.45 ± 0.10	<u>0.31</u> ± 0.15	0.70 ± 0.11	1.63 ± 0.09
MSX, exponential only	0.91 ± 0.13	<u>0.19</u> ± 0.11	11.51 ± 0.46	<u>0.32</u> ± 0.08	<u>0.82</u> ± 0.08
MSX-auto	0.24 ± 0.12	0.33 ± 0.13	0.30 ± 0.13	0.32 ± 0.08	0.74 ± 0.10

6.2 Public temporal point process benchmarks

We use the EasyTPP splits of Amazon, Retweet, Taxi, Taobao and StackOverflow and follow the protocol of Chang et al. (2024). The metric is the log-likelihood per scored event, with the first event of each sequence used only for conditioning, and models are selected on validation likelihood. Published numbers are taken from Chang et al. (2024). To check our evaluation code, we re-ran S2P2 on Taxi and obtained 0.524, against 0.522 in the paper. EPT-X reports the exact likelihood.

On Retweet, the largest dataset with 2.2 million training events, EPT-X improves on the best published model by more than 0.6 nats per event (Table 2). This result uses a 25-epoch budget imposed by compute, and the validation likelihood was still rising when training stopped. On Taobao (our 1.316 ± 0.005 against IntensityFree’s published 1.318 ± 0.017) and on StackOverflow (-2.166 ± 0.001 against S2P2’s -2.163 ± 0.009), the differences are within one published standard deviation.

On Taxi, EPT-X is 0.02 nats per event behind S2P2. A single-seed diagnostic run with the same test likelihood (0.502) has the same time term as our S2P2 re-run (0.733 against 0.735 nats per event), so the difference lies in the mark term. Amazon is different. Its gaps are concentrated in a narrow band (median 0.73, 99th percentile 0.80), and Erlang atoms of order at most 16 are too wide to fit this peak. With atoms of order up to 1024 on a finer grid, EPT-X gains more than 1.1 nats per event and moves into second place. Here our mark term is slightly better than S2P2’s (-1.89 against -1.90 on the re-run), and the remaining gap is in the time term. The linear MSX model comes close to the neural models on Taobao and Retweet, but it is far behind on Amazon and Taxi, whose dynamics are not self-exciting.

Tuning budgets. We selected the EPT-X configuration on validation data from a search over Erlang orders, grid size, encoder depth and width, an attention encoder, batch size and weight decay, with one seed per variant (Table 6 and Appendix E). The published baselines, in contrast, carry their authors’ own tuning, and our order-book baselines use the authors’ EasyTPP configurations with one seed each. The small margins on Taobao and StackOverflow should be read with this in mind. The margins on Retweet and on the order books are large compared with any seed-to-seed variation we observed.

Table 2: Public benchmarks: test log-likelihood per event (higher is better). Published numbers are from Chang et al. (2024) (five seeds each). Our rows give the mean $\pm$ s.d. over seeds (EPT-X: three to five seeds; other rows one seed), and the EPT-X configuration is selected per dataset on validation. Bold: best; underlined: second best; shaded: our models.

	Amazon	Retweet	Taxi	Taobao	StackOverflow
<i>Neural TPPs, published (Chang et al., 2024)</i>					
RMTTPP	−2.136	−7.098	0.346	1.003	−2.480
SAHP	−2.074	−6.708	0.298	1.168	−2.341
THP	−2.096	−6.659	0.372	0.790	−2.338
IntensityFree	0.496	−10.344	0.453	1.318	−2.233
MHP	−2.091	−6.564	0.370	0.636	−2.346
NHP	0.129	<u>−6.348</u>	<u>0.514</u>	1.157	−2.241
AttNHP	0.484	−6.499	0.493	1.259	−2.194
S2P2	0.781	−6.365	0.522	1.304	−2.163
<i>Classical Hawkes, our runs</i>					
ADM4	−2.260	−6.832	−0.315	1.184	−2.473
Conditional law	—	−8.317	−2.922	−63.919	−16.140
Hawkes EM (nonparam.)	−1.002	−6.655	−0.308	1.269	−2.370
<i>Ours</i>					
MSX ($R = 2$)	−1.943	−6.569	−0.214	1.299	−2.292
MSX-auto	−1.927	−6.568	−0.193	1.305	−2.292
EPT	−1.203	—	0.437	1.311	−2.186
EPT-X	<u>0.640</u> ± 0.001	−5.716 ± 0.024	0.503 ± 0.004	<u>1.316</u> ± 0.005	<u>−2.166</u> ± 0.001

Next-event prediction. Table 3 reports the two prediction metrics used by Chang et al. (2024). For the time of the next event, EPT-X has the lowest RMSE on Retweet (14,336s against 15,911s for NHP) and StackOverflow (1.013 against 1.014 for S2P2), and equals the best published RMSE to three decimals on Amazon (0.327), Taxi (0.281) and Taobao (0.125). For the mark of the next event, its accuracy is below that of S2P2 and NHP on four of the five datasets. This is the same split between time and mark that the likelihood decomposition shows on Taxi.

Table 3: Next-event prediction on the public benchmarks: time RMSE (top, lower is better; Retweet in seconds) and mark accuracy (bottom, %, higher is better). Published numbers from Chang et al. (2024), Table 2(b,c). EPT-X: mean $\pm$ s.d. over seeds of the validation-selected configuration. Bold: best; underlined: second best; shaded: ours.

	Amazon	Retweet	Taxi	Taobao	StackOverflow
<i>Published (Chang et al., 2024)</i>					
RMTTP	0.338	16488	0.283	0.126	1.049
SAHP	0.335	16102	0.290	0.126	1.031
THP	0.332	16268	0.285	0.125	1.033
IntensityFree	0.327	16625	0.362	<u>0.125</u>	1.340
MHP	0.329	16109	0.284	0.126	1.046
NHP	0.339	<u>15911</u>	0.282	0.126	1.019
AttNHP	2.656	16171	1.739	0.130	1.256
S2P2	<u>0.327</u>	15987	0.281	0.126	<u>1.014</u>
<i>Ours</i>					
EPT-X	0.327 $\pm$ 0.000	14336 $\pm$ 29	<u>0.281</u> $\pm$ 0.001	0.125 $\pm$ 0.000	1.013 $\pm$ 0.002

	Amazon	Retweet	Taxi	Taobao	StackOverflow
<i>Published (Chang et al., 2024)</i>					
RMTTP	30.8	53.4	91.4	60.9	45.6
SAHP	32.4	57.5	91.4	60.5	44.7
THP	34.6	60.2	91.4	60.0	46.6
IntensityFree	35.9	50.4	91.8	61.0	45.6
MHP	35.1	60.0	91.4	60.7	46.5
NHP	<u>39.4</u>	61.4	<u>92.9</u>	61.5	47.1
AttNHP	38.9	60.7	92.6	61.3	48.2
S2P2	40.7	<u>61.3</u>	93.1	61.1	<u>47.5</u>
<i>Ours</i>					
EPT-X	35.9 $\pm$ 0.1	58.8 $\pm$ 1.1	91.7 $\pm$ 0.1	<u>61.4</u> $\pm$ 0.1	45.5 $\pm$ 0.0

6.3 Limit-order-book event streams

We use the public LOBSTER (Huang & Polak, 2011) samples of AAPL, AMZN, GOOG, INTC and MSFT from 21 June 2012, with six event types, a chronological 60/20/20 split and sequences of 128 events. EasyTPP rejects tied timestamps, so we displace tied messages by a uniform offset in $(0, 100 \text{ ns})$, and every model is trained and scored on these same files. The neural baselines run in EasyTPP with the S2P2 authors’ configurations, with batches of 32 to 64 instead of 256 to fit an 8 GB GPU. Each is trained with a 100-epoch budget, and the strongest ones also with longer budgets. We report each at the budget with the best validation likelihood, so the longer budgets can only help them.

EPT-X is the best model on every stock (Table 4). It is ahead of MSX-auto, EPT and all seven neural baselines, and ahead of the best neural baseline by 0.23 to 0.45 nats per event.

One baseline needs a correction. EasyTPP’s IntensityFree replaces every gap below 10^{-5} s by 10^{-5} s before evaluating its log-normal mixture, which can then put an arbitrarily narrow component on that value. The result is not the likelihood of the observed gaps. On INTC and MSFT, 29% and 32% of the test gaps are this short (10 to 14% on the other stocks), and there the clamped model reports the highest numbers. Retrained and scored on the actual gaps with the same 1500-epoch budget, it loses 1.1 and 0.7 nats per event and falls below EPT-X, on INTC also below MSX. For scale, a denser MSX dictionary or sharper EPT-X atoms change the INTC validation likelihood by at most about 0.05. We report both versions, rank only the corrected one, and include the patch in the supplementary code.

Table 4: LOBSTER order books: test log-likelihood per event (higher is better; mean $\pm$ s.d. over seeds where several were run). Bold: best; underlined: second best; shaded: our models. *As shipped, IntensityFree clamps gaps at 10^{-5} s before scoring, so its score is not the likelihood of the observed data; it is greyed and not ranked. The IntensityFree row is retrained and scored on the actual gaps.

	AAPL	AMZN	GOOG	INTC	MSFT
<i>Classical Hawkes</i>					
Exp. kernel MLE	1.961	–	−0.311	–	–
Hawkes EM (nonparam.)	3.370	2.243	2.851	5.771	6.258
ADM4	1.961	1.067	1.306	4.787	5.211
Conditional law	−1.018	−17.814	−32.784	0.379	−1.270
<i>Neural TPPs (EasyTPP, our runs)</i>					
RMTTP	0.243	−0.686	−0.948	2.882	3.285
THP	−0.081	−0.948	−1.075	1.714	1.855
SAHP	1.777	0.462	−0.428	3.915	4.557
NHP	1.374	0.842	0.846	3.874	4.183
AttNHP	1.220	−0.168	0.004	3.497	3.771
S2P2	2.315	0.717	0.990	4.949	5.416
IntensityFree	3.108	2.103	2.726	5.713	6.377
IntensityFree* (as shipped)	3.260	1.921	2.679	6.790	7.070
<i>Ours</i>					
MSX ($R = 2$)	3.326	2.249	2.870	5.812	6.307
MSX-auto	3.361	<u>2.281</u>	<u>2.883</u>	5.814	6.310
EPT	<u>3.380</u> ± 0.008	2.229 ± 0.003	2.806 ± 0.005	<u>6.024</u> ± 0.007	<u>6.530</u> ± 0.005
EPT-X	3.534 ± 0.016	2.332 ± 0.005	2.992 ± 0.012	6.163 ± 0.002	6.666 ± 0.001

6.4 Ablations

Table 6 isolates the components of EPT-X on validation data.

- *Renewal channel.* Adding it improves EPT on every dataset.
- *Sharp atoms.* They help where the gap distribution has sharp peaks: by 1.1 nats per event on Amazon, where a finer grid helps further, and by 0.05 on INTC. Elsewhere they are neutral or slightly worse.
- *Encoder.* A causal Transformer encoder (Vaswani et al., 2017) in place of the GRU lowers the validation likelihood on all five datasets. A deeper encoder (three residual layers with LayerNorm and dropout), a wider one (128 units) and S2P2’s larger batch do not help either, so S2P2’s edge on Taxi is not a matter of encoder capacity or optimisation.
- *Hawkes backbone.* Without it, EPT reaches 2.84 nats per event on LOBSTER AAPL, below the classical MSX estimator.

Cost. Exactness is not free in our implementation (Table 5). On the same GPU, an EPT-X epoch on LOBSTER AAPL takes about four times as long as an S2P2 epoch, because our event recursion is a sequential loop while the Monte Carlo baselines vectorise over events. The exact compensator adds only a few closed-form terms per event, so a parallel-scan implementation of the Erlang recursion should remove most of this overhead.

Table 5: Training cost per epoch (including validation) and model size. EasyTPP baselines are timed over their full epoch budget; EPT-X stops early on validation.

Data	Model	Compensator	Device	s / epoch	Parameters
Taxi	EPT-X	exact	CPU (4 thr.)	15.4	146k
	S2P2	Monte Carlo	GPU	7.7	35k
LOBSTER AAPL	EPT-X	exact	GPU	22.0	76k
	S2P2	Monte Carlo	GPU	5.8	34k
	NHP	Monte Carlo	GPU	1.6	232k
	AttNHP	Monte Carlo	GPU	7.5	20k
	THP	Monte Carlo	GPU	0.4	51k
	IntensityFree	exact	GPU	0.4	7k

Table 6: EPT-X ablations: validation log-likelihood per event (higher is better). The EPT and EPT-X rows are the final runs (mean $\pm$ s.d. over seeds); the variants are single-seed selection runs with a 100-epoch budget. Bold: best; underlined: second best.

	Taxi	Taobao	StackOverflow	Amazon	INTC
<i>Validation log-likelihood per event</i>					
EPT (no renewal channel)	0.392	<u>1.200</u>	-2.186	-1.193	5.360 $\pm$ 0.007
EPT-X	0.466 $\pm$ 0.004	1.205 $\pm$ 0.002	-2.161 $\pm$ 0.003	-0.487 $\pm$ 0.000	5.463 $\pm$ 0.003
+ sharp atoms (orders to 1024)	<u>0.457</u>	1.194	<u>-2.173</u>	<u>0.604</u>	5.516
+ sharp atoms, 48 scales	0.455	1.177	-2.182	0.643	<u>5.511</u>
+ deep encoder (3 layers)	0.396	1.142	-2.264	-0.507	5.461
+ sharp atoms + deep encoder	0.398	1.152	-2.248	0.579	5.496
+ wide encoder (128 units)	0.431	1.188	-2.195	-0.489	5.459
+ attention encoder (causal Transformer)	0.415	1.169	-2.238	-0.531	5.416
batch 256 (S2P2 setting)	0.440	1.189	-2.442	-0.518	5.435

7 Application: how fast do markets absorb macroeconomic news?

Figure 2 summarises the steps from raw data to absorption times.

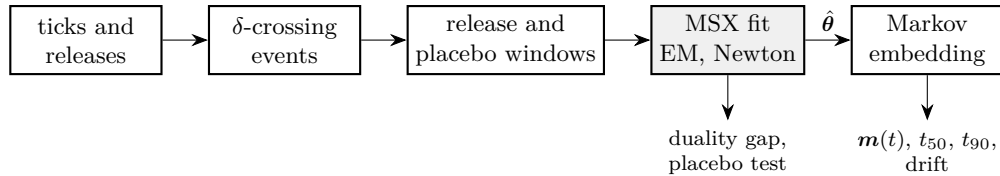


Figure 2: Measurement pipeline. Prices become δ -crossing events, which are grouped into release windows and matched placebo windows. A single certified MSX fit gives the duality gap and the placebo test, and its Markov embedding gives the response $\mathbf{m}(t)$, the absorption times and the expected price drift in closed form.

Releases. Our calendar covers 670 scheduled releases between January 2022 and August 2026. It contains CPI, nonfarm payrolls, PPI, retail sales, the GDP advance estimate and initial jobless claims, each dated by its first vintage in ALFRED, together with FOMC statements and press conferences, ECB decisions and press conferences, and Bank of Japan decisions. The surprise of a release is the difference between its first print and a real-time expectation, standardised and split into positive and negative parts. As expectations we use the Cleveland Fed inflation nowcast for CPI, the Atlanta Fed GDPNow forecast for GDP, and a statistical forecast from earlier first prints for the other releases.

Prices and events. We use millisecond quotes for EUR/USD, USD/JPY, gold and an S&P 500 contract for difference, and Binance futures trades for bitcoin and ether. Each price series is turned into events by

δ -crossing: an upward (downward) event is recorded whenever the log price has risen (fallen) by δ since the previous event. We choose δ per asset, between 0.25 and 3 basis points, so that the median rate is about 0.2 events per second. Each asset thus contributes two event types, and the twelve types together form the dimensions of the model. Windows run from 30 minutes before a release to 60 minutes after it, preceded by 30 minutes of history that is used for the states but not scored.

Placebo windows. Each release window is paired with a placebo window at the same clock time on the same weekday, one to six weeks away, chosen so that no modelled release and none of sixteen second-tier US releases falls within three hours. Placebo windows contain pseudo-releases of their own type, so the placebo test is part of the likelihood rather than a separate regression. The pooled model has 732 windows (430 release and 302 placebo) with 7.3 million scored events, and its certificate bounds the distance to the optimum by 0.0054 nats per dimension.

Endogeneity. The spectral radius of the branching matrix is $\rho(G) = 0.87$, so roughly seven in every eight price events are triggered by earlier price events rather than directly by news. Fitted year by year, $\rho(G)$ declines from 2022 to 2026 (0.91, 0.89, 0.85, 0.86, 0.85), and the slowest echo mode decays in about 12 minutes.

Falsification. Under the null hypothesis of no effect, a nonnegative news kernel still has a positive expected mass, because estimates cannot fall below zero. A comparison with release-free placebo windows is therefore essential. With the ℓ_1 level chosen by held-out likelihood, 5 of the 11 release types exceed their placebo by at least 50% (Figure 3): CPI (8.5 $\times$), payrolls (6.7 $\times$), PPI (4.0 $\times$), the FOMC statement (3.9 $\times$) and retail sales (1.9 $\times$). Jobless claims, the ECB and BoJ decisions and the press conferences show no direct effect beyond their placebos. Two earlier specifications failed this test, and we report them in Appendix B.

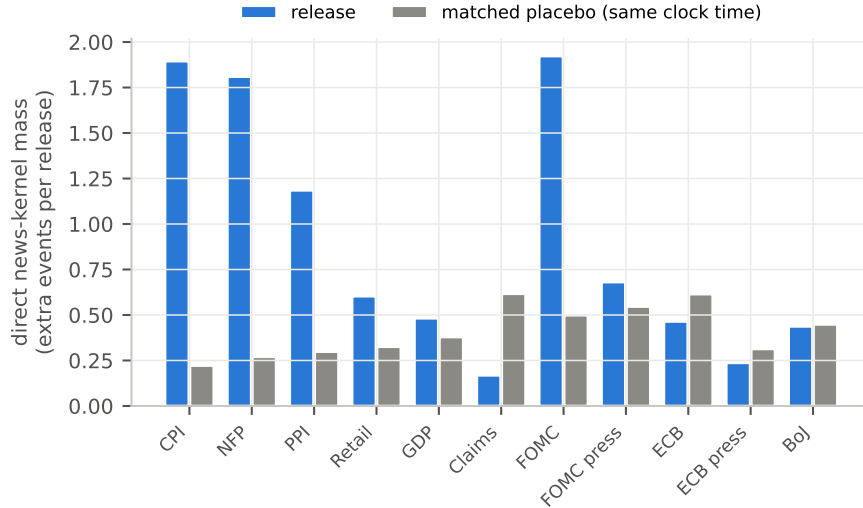


Figure 3: Placebo test. Direct news-kernel mass, the expected number of extra events per release summed over all assets, for each release type (blue) and for its matched placebo at the same clock time on a release-free day (grey). A type passes if its mass exceeds the placebo mass by at least 50%.

Echo-corrected absorption. For the releases that pass the placebo test, the direct news kernel is absorbed almost immediately: its median t_{50} is 1.1 s. The echoes stretch the response considerably. Half of the extra activity arrives within 16 to 137 s, and ninety percent within 5 to 11 minutes (Figure 4 and Table 7). The “news half-life” $\ln 2/\beta$ therefore understates absorption time by one to two orders of magnitude. Because positive and negative surprises enter the news kernels separately, the model also gives expected price paths after a release (Figure 5). A higher-than-expected CPI print lowers bitcoin and ether, and a lower one lifts them.

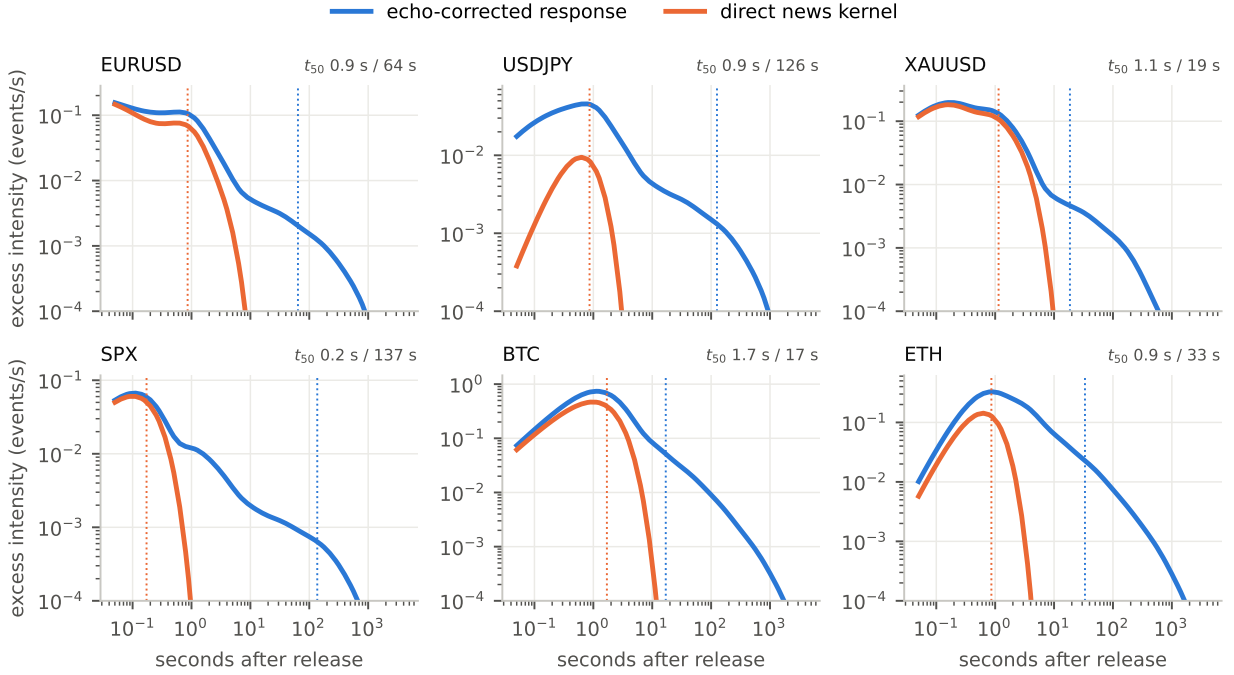


Figure 4: Response of each asset to a CPI release, in excess events per second (both axes logarithmic). Orange: direct news kernel. Blue: echo-corrected response $m(t)$, which adds the events triggered by earlier events. Dotted lines and panel titles give t_{50} , direct / echo-corrected.

Table 7: Echo-corrected absorption times from the pooled model: t_{50} in seconds / t_{90} in minutes after the release. *Release types whose news-kernel mass does not exceed their placebo by 50% (Figure 3).

Release	EURUSD	USDJPY	Gold	S&P 500	BTC	ETH
CPI	64 / 9.1	126 / 11.0	19 / 4.8	137 / 11.0	17 / 6.1	33 / 7.8
NFP	94 / 10.1	109 / 10.7	16 / 5.2	80 / 8.8	27 / 7.5	24 / 7.3
PPI	120 / 10.8	94 / 10.3	40 / 6.4	79 / 8.6	23 / 7.2	39 / 8.7
FOMC	89 / 10.1	76 / 9.8	82 / 9.0	77 / 8.9	41 / 9.2	36 / 9.1
Retail	59 / 9.1	77 / 9.8	87 / 9.2	114 / 10.4	26 / 7.6	26 / 7.8
GDP*	95 / 10.0	81 / 9.6	26 / 5.1	145 / 11.2	27 / 7.3	27 / 7.5
FOMC press conf.*	88 / 10.0	111 / 10.9	60 / 8.1	59 / 8.0	45 / 9.5	52 / 10.0
Claims*	124 / 11.0	56 / 9.1	87 / 9.3	81 / 9.1	36 / 9.0	40 / 9.4
ECB*	61 / 9.2	73 / 9.6	88 / 9.0	123 / 10.8	37 / 9.2	42 / 9.6
ECB press conf.*	137 / 11.3	90 / 10.2	38 / 6.5	63 / 8.2	25 / 7.3	32 / 8.0
BoJ*	93 / 10.2	80 / 9.9	45 / 7.3	73 / 8.8	35 / 8.5	37 / 8.8

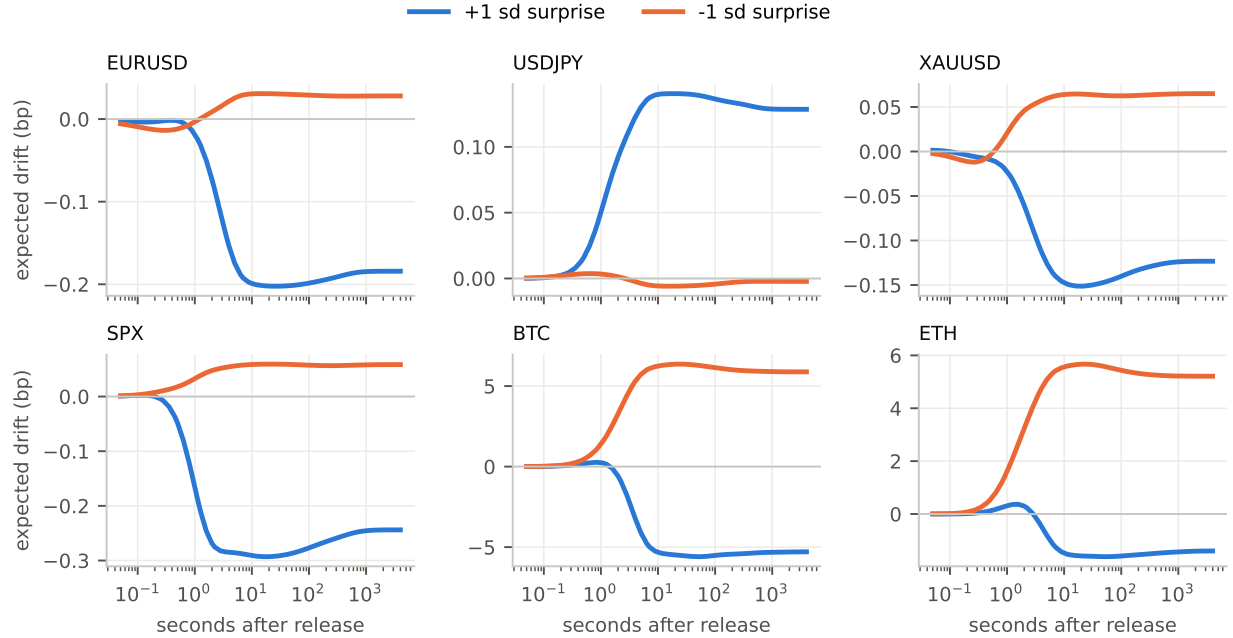


Figure 5: Expected cumulative price change, in basis points, after a CPI surprise of +1 (blue) or -1 (orange) standard deviation, from the pooled model. Time axis logarithmic.

Goodness of fit. Time-rescaled residuals (Brown et al., 2002) have Kolmogorov–Smirnov statistics of at most 0.044 against $\text{Exp}(1)$; the per-window excess-dispersion test rejects in 26–55% of windows, compared with 42% (power-law kernel) and 93% (double exponential) reported for FX data by Rambaldi et al. (2015).

Forecasting. Out of sample, closed-form MSX forecasts of post-release activity improve on climatology but not on regressions fitted directly to that target, whereas its drift forecasts are more informative than the best baseline (Appendix D).

8 Discussion and limitations

Writing every kernel as a nonnegative combination of phase-type densities makes Hawkes estimation a convex problem whose solution can be certified, and it turns the response to a shock into a matrix exponential. Adding a neural history encoder and a mixture-of-Erlang renewal channel to the same state space gives a model whose likelihood is still exact. This model is competitive with the strongest neural temporal point processes, and better than them on the order books and on Retweet.

We draw three conclusions. First, the speed at which news is absorbed is not a property of the news kernel alone. The market’s own echoes lengthen it by one to two orders of magnitude, and the correction can be computed in closed form. Second, falsification tests are worth building in. Two of our earlier specifications produced plausible news effects that the placebo test showed to be artefacts. Third, benchmark comparisons should check that every model is scored on the same observed data. A single clamp inside one implementation was enough to reverse a ranking.

Limitations. For payrolls, claims, PPI and retail sales our expectations are statistical forecasts rather than survey consensus, and the FX data have a gap in the first half of 2023. The order-book evaluation uses the public LOBSTER samples, which cover one trading day per stock. On the order books, the neural baselines were trained with smaller batches than their authors used, to fit an 8 GB GPU. On the public benchmarks we compare with published numbers and have checked our evaluation with one re-run. On Retweet, EPT-X was trained for 25 epochs and its validation likelihood was still rising, so the reported

margin is, if anything, conservative. EPT-X does not reach the best published likelihood on Taxi, where the gap is in mark prediction, or on Amazon, where it is in the timing density. Finally, the MSX forecasts of post-release activity are beaten by regressions fitted directly to that target.

Broader Impact Statement

This work models market event streams with public or licensable data and makes no use of personal data. Better models of how quickly prices absorb public information can inform market monitoring and the timing of disclosures, and exact likelihoods make benchmark comparisons more reliable. Faster or more accurate event forecasts could also be used in trading; the methods described are standard statistical tools and do not by themselves confer an informational advantage. We release code and patches so that the reported numbers, including the corrected baseline, can be reproduced.

Reproducibility Statement

All result tables and every number quoted in the text are generated by scripts from the logged outputs of the experiments; the supplementary code contains the models, the experiment and table scripts, the EasyTPP patches (tie handling and the IntensityFree clamp), configuration files and unit tests (exact compensators against numerical quadrature, causality of the encoders, nesting of EPT-X in EPT). Public datasets are used through their standard splits; the market data are licensed from their providers and are not redistributed.

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A Proofs

Proof of Proposition 2. We use the notation of the proposition: X has rows $\mathbf{x}_n^\top = \mathbf{x}(t_n)^\top$, $\lambda_n = \mathbf{x}_n^\top \boldsymbol{\theta}$, and $\ell(\boldsymbol{\theta}) = \sum_{n=1}^N \log \lambda_n - (\boldsymbol{\Phi} + \boldsymbol{\gamma})^\top \boldsymbol{\theta}$.

Step 1 (an upper bound for every feasible point). For $u, v > 0$, Fenchel’s inequality gives

$$\log u \leq v u - 1 - \log v.$$

Take any vector $\mathbf{v} = (v_1, \dots, v_N)$ with positive entries such that $X^\top \mathbf{v} \leq \boldsymbol{\Phi} + \boldsymbol{\gamma}$ componentwise. Applying the inequality with $u = \lambda_n$ and $v = v_n$ and summing over n ,

$$\sum_n \log \lambda_n \leq \sum_n v_n \lambda_n - N - \sum_n \log v_n = \boldsymbol{\theta}^\top X^\top \mathbf{v} - N - \sum_n \log v_n.$$

Since $\boldsymbol{\theta} \geq 0$ and $X^\top \mathbf{v} \leq \boldsymbol{\Phi} + \boldsymbol{\gamma}$, we have $\boldsymbol{\theta}^\top X^\top \mathbf{v} \leq \boldsymbol{\theta}^\top (\boldsymbol{\Phi} + \boldsymbol{\gamma})$. Subtracting $(\boldsymbol{\Phi} + \boldsymbol{\gamma})^\top \boldsymbol{\theta}$ from both sides therefore gives

$$\ell(\boldsymbol{\theta}) \leq D(\mathbf{v}) := -N - \sum_n \log v_n.$$

This holds for every feasible $\boldsymbol{\theta}$, in particular for the maximiser, so $\ell^* \leq D(\mathbf{v})$.

Step 2 (a computable choice of $\mathbf{v}$). Let $\boldsymbol{\theta}$ be the current fit and set $v_n = 1/(s\lambda_n)$, with $s = \max_p (\sum_n x_{np}/\lambda_n)/(\Phi_p + \gamma_p)$. Then $(X^\top \mathbf{v})_p = \frac{1}{s} \sum_n x_{np}/\lambda_n \leq \Phi_p + \gamma_p$ by the definition of s , so $\mathbf{v}$ is admissible in Step 1. For this choice,

$$D(\mathbf{v}) = -N + N \log s + \sum_n \log \lambda_n, \quad \text{hence} \quad \ell^* - \ell(\boldsymbol{\theta}) \leq D(\mathbf{v}) - \ell(\boldsymbol{\theta}) = N \log s + \boldsymbol{\theta}^\top (\boldsymbol{\Phi} + \boldsymbol{\gamma}) - N.$$

Step 3 (the bound vanishes at the optimum). The Karush–Kuhn–Tucker conditions of the maximisation state that $\sum_n x_{np}/\lambda_n = \Phi_p + \gamma_p$ when $\theta_p > 0$ and $\sum_n x_{np}/\lambda_n \leq \Phi_p + \gamma_p$ when $\theta_p = 0$. Hence $s = 1$, so $N \log s = 0$. Moreover,

$$\boldsymbol{\theta}^\top (\boldsymbol{\Phi} + \boldsymbol{\gamma}) = \sum_p \theta_p \sum_n \frac{x_{np}}{\lambda_n} = \sum_n \frac{\mathbf{x}_n^\top \boldsymbol{\theta}}{\lambda_n} = N,$$

so the second term is also zero. □

B Specification history of the application

We report the specifications that failed the built-in falsification test. The final specification was chosen because it passed this test, not because of the results it produced.

Version 1. This version used news kernels with time scales up to 30 minutes and Erlang order 3, a constant baseline in each window and time-of-day effects in UTC. It also allowed tied events: a trade that crossed several δ levels produced several events with the same timestamp. The single placebo type absorbed 169 extra events per pseudo-release, which is comparable to real releases, and the residuals of the crypto series had Kolmogorov–Smirnov statistics up to 0.58. Two causes explain this. News kernels longer than the post-release window absorbed slow trends within the window, and tied events violate the assumption that events occur one at a time.

Version 2. This version kept one event per timestamp, with the number of crossed levels as a mark, restricted news kernels to time scales between 0.1 and 300 s, used a linear baseline within each window and modelled seasonality in New York time. The placebo mass fell to 18.6 but was still significant. Placebo mass appeared at every release time of day, so unmodelled 08:30 releases could not be the whole explanation.

Version 3 (final). Placebo days now also avoid sixteen second-tier US releases. The news kernels carry an ℓ_1 penalty, weighted by exposure, whose level is chosen by the likelihood of held-out windows (Table 8). Without the penalty, nonnegative estimates inflate release and placebo masses equally. The held-out likelihood peaks at $\ell_1 = 0.8$, which is also where release and placebo masses separate.

Table 8: Choice of the ℓ_1 penalty on the news kernels. Held-out log-likelihood on the 2024 windows relative to the best level (bold), and the mean news-kernel mass of release and placebo windows.

ℓ_1 level	held-out LL (rel.)	mean release mass	mean placebo mass
0.2	−93.7	10.79	6.23
0.4	−27.1	5.20	2.89
0.8	0.0	1.81	1.03
1.6	−20.5	0.44	0.37

C Synthetic recovery: all metrics

Table 9: Synthetic recovery, all metrics (mean $\pm$ s.d. over five seeds): relative error of the branching matrix and of its spectral radius, kernel L^1 error and held-out deficit dLL (10^{-3} nats/event), all lower is better, and edge AUC (higher is better). AUC needs absent true edges, so it is defined only for S4 and S5 (n/a for the complete graphs S1–S3). “–”: not produced by the method; NPHC estimates only the branching matrix. Missing rows are fits that did not converge. Bold: best; underlined: second best within each scenario; shaded: ours.

	G error	ρ error	kernel L^1	dLL (10^{-3})	edge AUC
<i>S1: exponential</i>					
Exp. kernel MLE	0.018 $\pm$ 0.007	0.005 $\pm$ 0.003	0.018 $\pm$ 0.008	0.05 $\pm$ 0.05	n/a
Sum-of-exp. MLE	0.065 $\pm$ 0.022	0.030 $\pm$ 0.013	0.165 $\pm$ 0.023	0.92 $\pm$ 0.13	n/a
ADM4	<u>0.018</u> $\pm$ 0.007	<u>0.005</u> $\pm$ 0.003	<u>0.018</u> $\pm$ 0.008	<u>0.05</u> $\pm$ 0.05	n/a
Hawkes EM (nonparam.)	0.035 $\pm$ 0.015	0.017 $\pm$ 0.007	0.173 $\pm$ 0.010	1.02 $\pm$ 0.07	n/a
NPHC (cumulants)	0.247 $\pm$ 0.087	0.032 $\pm$ 0.024	–	–	n/a
Conditional law	0.120 $\pm$ 0.054	0.017 $\pm$ 0.016	0.193 $\pm$ 0.022	2.84 $\pm$ 1.03	n/a
MSX-auto	0.054 $\pm$ 0.032	0.020 $\pm$ 0.018	0.093 $\pm$ 0.035	0.24 $\pm$ 0.12	n/a
MSX ($R = 2$)	0.056 $\pm$ 0.032	0.019 $\pm$ 0.018	0.092 $\pm$ 0.035	0.23 $\pm$ 0.12	n/a
MSX ($R = 4$)	0.056 $\pm$ 0.035	0.023 $\pm$ 0.018	0.105 $\pm$ 0.034	0.28 $\pm$ 0.15	n/a
MSX, exponential only	0.055 $\pm$ 0.025	0.026 $\pm$ 0.015	0.157 $\pm$ 0.022	0.91 $\pm$ 0.13	n/a
<i>S2: power law</i>					
Exp. kernel MLE	0.299 $\pm$ 0.011	0.219 $\pm$ 0.007	0.465 $\pm$ 0.003	26.05 $\pm$ 0.50	n/a
Sum-of-exp. MLE	<u>0.057</u> $\pm$ 0.012	0.016 $\pm$ 0.007	0.094 $\pm$ 0.011	0.18 $\pm$ 0.12	n/a
ADM4	0.299 $\pm$ 0.011	0.219 $\pm$ 0.007	0.465 $\pm$ 0.003	26.05 $\pm$ 0.50	n/a
Hawkes EM (nonparam.)	0.057 $\pm$ 0.014	0.037 $\pm$ 0.008	0.188 $\pm$ 0.006	1.73 $\pm$ 0.21	n/a
NPHC (cumulants)	0.153 $\pm$ 0.154	0.041 $\pm$ 0.047	–	–	n/a
Conditional law	0.102 $\pm$ 0.067	0.029 $\pm$ 0.019	0.291 $\pm$ 0.028	4.91 $\pm$ 0.59	n/a
MSX-auto	0.064 $\pm$ 0.029	0.037 $\pm$ 0.029	0.108 $\pm$ 0.022	0.33 $\pm$ 0.13	n/a
MSX ($R = 2$)	0.057 $\pm$ 0.021	<u>0.015</u> $\pm$ 0.010	0.114 $\pm$ 0.015	0.32 $\pm$ 0.12	n/a
MSX ($R = 4$)	0.062 $\pm$ 0.014	0.016 $\pm$ 0.006	0.131 $\pm$ 0.017	0.45 $\pm$ 0.10	n/a
MSX, exponential only	0.054 $\pm$ 0.022	0.014 $\pm$ 0.010	<u>0.097</u> $\pm$ 0.011	<u>0.19</u> $\pm$ 0.11	n/a
<i>S3: hump</i>					
Exp. kernel MLE	0.166 $\pm$ 0.008	0.053 $\pm$ 0.004	0.801 $\pm$ 0.004	101.47 $\pm$ 1.81	n/a
Sum-of-exp. MLE	0.038 $\pm$ 0.011	0.018 $\pm$ 0.005	0.419 $\pm$ 0.005	11.51 $\pm$ 0.46	n/a
ADM4	0.166 $\pm$ 0.008	0.053 $\pm$ 0.004	0.801 $\pm$ 0.004	101.47 $\pm$ 1.81	n/a
Hawkes EM (nonparam.)	0.024 $\pm$ 0.011	0.010 $\pm$ 0.004	0.187 $\pm$ 0.008	2.24 $\pm$ 0.12	n/a
NPHC (cumulants)	0.240 $\pm$ 0.096	0.045 $\pm$ 0.048	–	–	n/a
Conditional law	0.149 $\pm$ 0.062	0.017 $\pm$ 0.010	0.319 $\pm$ 0.050	10.66 $\pm$ 2.13	n/a
MSX-auto	<u>0.032</u> $\pm$ 0.016	0.016 $\pm$ 0.007	<u>0.080</u> $\pm$ 0.013	0.30 $\pm$ 0.13	n/a
MSX ($R = 2$)	0.069 $\pm$ 0.009	0.037 $\pm$ 0.004	0.347 $\pm$ 0.007	5.90 $\pm$ 0.36	n/a
MSX ($R = 4$)	0.038 $\pm$ 0.010	0.017 $\pm$ 0.005	0.074 $\pm$ 0.010	<u>0.31</u> $\pm$ 0.15	n/a
MSX, exponential only	0.034 $\pm$ 0.009	<u>0.015</u> $\pm$ 0.004	0.416 $\pm$ 0.008	11.51 $\pm$ 0.46	n/a
<i>S4: multiscale network</i>					
Exp. kernel MLE	0.173 $\pm$ 0.041	0.095 $\pm$ 0.025	1.635 $\pm$ 0.040	711.28 $\pm$ 53.09	1.000 $\pm$ 0.000
ADM4	0.399 $\pm$ 0.003	0.255 $\pm$ 0.001	0.410 $\pm$ 0.001	3.13 $\pm$ 0.25	1.000 $\pm$ 0.000
Hawkes EM (nonparam.)	0.153 $\pm$ 0.012	0.045 $\pm$ 0.010	0.358 $\pm$ 0.012	4.55 $\pm$ 0.29	1.000 $\pm$ 0.000
NPHC (cumulants)	1.030 $\pm$ 0.304	0.211 $\pm$ 0.186	–	–	0.588 $\pm$ 0.155
Conditional law	0.464 $\pm$ 0.286	0.139 $\pm$ 0.050	1.059 $\pm$ 0.074	114.94 $\pm$ 1.20	0.831 $\pm$ 0.087
MSX-auto	0.084 $\pm$ 0.014	<u>0.016</u> $\pm$ 0.006	0.128 $\pm$ 0.026	0.32 $\pm$ 0.08	1.000 $\pm$ 0.000
MSX ($R = 2$)	0.093 $\pm$ 0.015	0.021 $\pm$ 0.005	0.173 $\pm$ 0.024	0.54 $\pm$ 0.12	1.000 $\pm$ 0.000
MSX ($R = 4$)	0.101 $\pm$ 0.019	0.027 $\pm$ 0.007	0.207 $\pm$ 0.035	0.70 $\pm$ 0.11	1.000 $\pm$ 0.000
MSX, exponential only	<u>0.084</u> $\pm$ 0.014	0.016 $\pm$ 0.006	<u>0.128</u> $\pm$ 0.026	<u>0.32</u> $\pm$ 0.08	1.000 $\pm$ 0.000
<i>S5: 10-dim. power-law network</i>					
Exp. kernel MLE	0.337 $\pm$ 0.153	0.103 $\pm$ 0.032	1.302 $\pm$ 0.897	426.80 $\pm$ 368.57	0.977 $\pm$ 0.050
ADM4	0.167 $\pm$ 0.003	0.096 $\pm$ 0.003	0.319 $\pm$ 0.002	26.51 $\pm$ 0.29	1.000 $\pm$ 0.000
Hawkes EM (nonparam.)	<u>0.085</u> $\pm$ 0.008	0.045 $\pm$ 0.008	0.280 $\pm$ 0.010	5.23 $\pm$ 0.25	1.000 $\pm$ 0.000
NPHC (cumulants)	1.348 $\pm$ 0.477	0.520 $\pm$ 0.372	–	–	0.672 $\pm$ 0.042
Conditional law	0.461 $\pm$ 0.047	0.037 $\pm$ 0.026	0.641 $\pm$ 0.025	78.64 $\pm$ 4.79	0.950 $\pm$ 0.033
MSX-auto	0.045 $\pm$ 0.004	0.015 $\pm$ 0.005	0.079 $\pm$ 0.005	0.74 $\pm$ 0.10	1.000 $\pm$ 0.000
MSX ($R = 2$)	0.111 $\pm$ 0.013	0.040 $\pm$ 0.013	0.180 $\pm$ 0.017	1.19 $\pm$ 0.07	1.000 $\pm$ 0.000
MSX ($R = 4$)	0.122 $\pm$ 0.008	0.051 $\pm$ 0.013	0.218 $\pm$ 0.015	1.63 $\pm$ 0.09	<u>1.000</u> $\pm$ 0.000
MSX, exponential only	0.099 $\pm$ 0.014	<u>0.030</u> $\pm$ 0.014	<u>0.145</u> $\pm$ 0.017	<u>0.82</u> $\pm$ 0.08	1.000 $\pm$ 0.000

D Forecasting the post-release burst

On the 163 releases of 2025–2026, an MSX model fitted on 2022–2024 forecasts the number of δ -crossings in the first 1, 5 and 15 minutes in closed form (Table 10). It improves on same-type climatology at every horizon but is beaten by the event-study OLS and HAR regressions, which are fitted directly to this target. For the direction of the move the structural model is informative: the correlation between its forecast and the

realised one-minute drift is 0.17, against 0.06 for the best baseline (mean drift by release type and surprise sign).

Table 10: Post-release activity forecasts on the 163 test releases of 2025–2026, from models fitted on 2022–2024. N is the number of δ -crossings after the release; QLIKE scores the forecast of realised variance. Mean over assets; lower is better. Bold: best; underlined: second best; shaded: our model.

Method	RMSE of $\log(1 + N)$			QLIKE of realised variance		
	1 min	5 min	15 min	1 min	5 min	15 min
Climatology	1.440	1.308	1.199	1.263	0.916	0.701
Event-study OLS	1.140	0.925	0.776	2.552	0.580	0.372
HAR	<u>1.217</u>	<u>0.985</u>	<u>0.820</u>	<u>1.467</u>	<u>0.762</u>	<u>0.468</u>
MSX closed form	1.322	1.049	0.955	2.913	1.036	0.518

E Experimental details

MSX in the application.

- Endogenous kernels: 10 rates from 5 ms to 600 s, Erlang order 2.
- News kernels: 8 rates from 0.1 to 300 s, Erlang order 3, marks $(1, z^+, z^-)$.
- Anticipation kernels: 4 rates from 10 to 300 s.
- Baseline: a linear trend within each window plus 96 time-of-day hat functions.

MSX on the benchmarks. MSX uses one baseline shared by all sequences and a grid of 12 rates from the 1st percentile of the training gaps to five times their 99.9th percentile. MSX-auto chooses the Erlang order from $\{1, 2, 3, 4\}$ and the ℓ_1 level from $\{0, 0.01, 0.05\}$ on the validation split.

EPT and EPT-X.

- Encoder: a GRU with 64 units. Latent state: 4 channels of 8 learnable rates. The number of Erlang phases was chosen on validation from $\{2, 4\}$, together with the hidden size from $\{64, 128\}$, on Taxi, Taobao and StackOverflow.
- Hawkes backbone: 10 rates with 2 phases each.
- Optimisation: AdamW (Loshchilov & Hutter, 2019) with learning rate 10^{-2} , 1% linear warm-up and cosine decay, gradient clipping at 1, and batches of 64 that are halved automatically if the GPU runs out of memory. Training stops when the validation likelihood has not improved for 40 epochs, within a budget of 300 epochs (25 epochs with batches of 32 on Retweet).
- Renewal channel: 24 mean gaps spaced logarithmically from half the 0.1% quantile of the training gaps to five times their 99.9% quantile, with orders $\{1, 4, 16\}$. The mark distributions start at the empirical mark frequencies.
- Selected on validation, the LOBSTER streams and Amazon add orders up to 1024 and a history-dependent shift of the atoms. Amazon also uses a finer grid: the single-seed validation log-likelihood was 0.604, 0.643, 0.653 and 0.649 with 24, 48, 96 and 192 scales, so we use 96.

Baselines. The neural baselines run in EasyTPP with the authors’ configuration files. The classical Hawkes estimators use the `tick` library (Bacry et al., 2018), with positive-constrained accelerated gradient descent, tolerance 10^{-10} and decay grids matched to MSX. A short patch, included in the supplementary code, makes the IntensityFree clamp configurable.

Compute. All experiments ran on one laptop with an 8 GB GPU and a 28-core CPU, with each CPU job limited to four threads. EPT-X runs about as fast on the CPU as on the GPU, because its event recursion is sequential.